

I. Trinomials

$$ax^2 + bx + c = 0$$

Continued from page 1. $6x^2 - 11x - 10 = 0$

7. Test Option A. Multiply the outer pair and the inner pair.

Those two products have to combine to make the middle term, $-11x$. Leave the signs out for now.

$$\begin{array}{c} 5 \cdot 2x = 10x \\ \swarrow \quad \searrow \\ (3x \quad \boxed{5}) (2x \quad \boxed{2}) \\ \swarrow \quad \searrow \\ 3x \cdot 2 = 6x \end{array}$$

✗ 6 and 10 make 16 or 4. Neither one is 11, so this arrangement is wrong — swap the 5 and the 2.

8. Test Option B the same way.

$$\begin{array}{c} 2 \cdot 2x = 4x \\ \swarrow \quad \searrow \\ (3x \quad \boxed{2}) (2x \quad \boxed{5}) \\ \swarrow \quad \searrow \\ 3x \cdot 5 = 15x \end{array}$$

✓ 15 and 4 make 19 or 11. That 11 is exactly what we need, so the numbers are in the right spots.

9. Now decide the signs.

We need $-11x$, so the bigger product has to be the negative one.

$$\begin{array}{r} -15x \\ + \quad 4x \\ \hline -11x \end{array}$$

The 15 came from $3x \cdot 5$, so the **5** is the one that turns negative.

$$(3x + 2)(2x - 5)$$

Check the last term: $2 \cdot (-5) = -10$ ✓

10. Set each factor equal to zero and solve.

If two things multiply to zero, one of them has to be zero.

$$(3x + 2)(2x - 5) = 0$$

$$3x + 2 = 0$$

$$3x = -2$$

$$\boxed{x = -\frac{2}{3}}$$

$$2x - 5 = 0$$

$$2x = 5$$

$$\boxed{x = \frac{5}{2}}$$

Dividing by 2 back in step 1 did not change these answers.